

# UK Road Safety Collisions 2024

## 1. INTRODUCTION

This report presents an analysis of road safety collision data for 2024, sourced from the Department for Transport (DfT). All statistics contained within this report relate exclusively to Great Britain. The report provides an overview of key collision metrics across Great Britain, including total collisions, total casualties, and Killed or Seriously Injured (KSI) figures. It explores the data by collision severity, road type, urban or rural area, and lighting conditions, enabling targeted analysis of road safety patterns and risk factors at a national level.

## 2. TARGET AUDIENCE

This dashboard is designed for road safety policy analysts working within local authorities and the Department for Transport (DfT). The data is directly relevant to this audience because it enables the identification of high-risk collision patterns by road type, severity, geography, lighting, and area that are needed to prioritise infrastructure investment, enforcement strategies, and casualty reduction targets under the Government's Road Safety Framework.

## 3. REPORT TYPE RATIONALE

This analytical report provides the primary users, road safety analysts, who need to interrogate the data in cross-filtering by severity, location type, and light conditions to reveal risk factors. The Power BI dashboard's slicers support this: toggling between Fatal, Serious, and Slight; switching between Urban and Rural; and layering light condition filters all produce materially different KPI readings, which is the behaviour an analyst needs when building an evidence base for policy decisions.

## 4. METHODS

DAX measures were created with Business Intelligence (BI) to gain meaningful metrics from the raw collision data. Total Collisions counts all records in the Collisions table, while Total Casualties sum up the number of casualties per collision. Fatal Collisions filters the dataset to count only collisions where severity is recorded as fatal. KSI Count extends this by combining fatal and serious collisions into a single measure. KSI Rate % divides KSI Count by Total Collisions to express the proportion of collisions resulting in killed or seriously injured casualties as a percentage, updating in response to any slicer selection. Average Casualties per Collision divides total casualties by total collisions to indicate the mean harm caused per incident. Total Casualties by Type counts rows in the Casualties table to show how many casualties occurred per casualty category (pedestrians, cyclists, motorcyclists, etc.), enabling the dashboard to display which road user groups are most at risk. Together, these measures shift the dashboard beyond simple counts towards comparative analysis.

## 5. INSIGHTS FROM THE VISUALS

### 5.1 Scale and KSI burden

Across 2024, the dashboard records **101,000 total collisions** generating **128,000 total casualties**. Of these, **25,000 meet the Killed or Seriously Injured (KSI) threshold**, representing a KSI rate of **24.84%**, meaning roughly one in four collisions results in a KSI outcome. It is optimal for policy analysts to compare this rate

against the DfT's rolling five-year average to assess whether 2024 represents progress or deterioration.

### 5.2 Single carriageways are the dominant collision environment

The "Collisions by Road Type" chart shows that **single carriageways account for 74,000 collisions (73% of the total)**, exceeding dual carriageways at 15,000 and roundabouts at 6,000. This concentration is not surprising given the mix of speeds, opposing traffic, and limited separation on single-lane A and B roads, but the magnitude reinforces the case for targeted interventions such as average speed cameras, lane separation, and junction upgrades on this road class.

### 5.3 Rural roads are disproportionately deadly

Filtering by area type reveals a stark disparity in KSI rates: **Urban collisions (67,000; 66% of the total)** carry a KSI rate of **22.46%**, while **Rural collisions (34,000; 34% of the total)** carry a significantly higher KSI rate of **29.60%**. Rural roads produce a third of the volume but generate a substantially higher proportion of KSI casualties, consistent with the higher average speeds and longer emergency response times on rural networks. In other words, rural safety investment needs greater weighting.

### 5.4 The monthly trend shows two distinct collision peaks

The Monthly Collisions Trend line chart shows a noticeable low point in **February** (approximately 7,500 collisions), a **primary peak in June** (approaching 9,000), a secondary peak in **October-November** (again near 9,000), and a notable **decline into December** (approximately 8,000). The summer peak likely reflects higher traffic volumes and increased activity, while the autumn secondary peak may reflect deteriorating light and weather conditions.

### 5.5 Car occupants and vulnerable road users dominate casualty counts

The "Casualties by Type" chart shows **car occupants at 68,000 casualties**, more than all other user types combined, followed by pedestrians (19,000), motorcyclists (16,000), and cyclists (16,000). The figures for motorcyclists and cyclists are notable: both represent relatively small shares of total traffic, yet each contributes approximately the same casualty volume, implying very high casualty rates per journey for these groups.

### 5.6 The severity split points to a large "hidden" serious injury problem

Of the 101,000 collisions, **76,000 are classified as Slight (75%)**, **24,000 as Serious (24%)**, and **2,000 as Fatal (2%)**. The serious injury cohort is twelve times larger than the fatal cohort.

### 5.7 A clear pattern emerges when toggling the light conditions slicer

"Dark (no light)" has a KSI rate of 34.63%, nearly 10 percent higher than daylight. Despite representing only 6% of all collisions, this condition is by far the deadliest ([Table 1](#)). The monthly trend for "Dark (no light)" also shows a clear winter peak (highest in

December/November), which makes sense as nights get longer. “Dark (lights lit)” collisions also peak in winter, the opposite of the overall trend, which peaks in summer/autumn.

Table 1

| Light Condition        | Collisions     | KSI Rate               |
|------------------------|----------------|------------------------|
| Daylight               | 72K (71%)      | 23.79%                 |
| Dark (lights lit)      | 21K (21%)      | 25.91%                 |
| Dark (no light)        | 6K (6%)        | 34.63%(most dangerous) |
| Dark(lights ukn/unlit) | small residual | –                      |

6. CHALLENGES

The 2024 STATS19 release uses numeric codes across multiple fields, requiring manual replacement with readable labels in Power Query. Casualty type codes had to be replaced in descending order to prevent partial string matches from corrupting shorter codes. For example, replacing 1 before 11 produced values like carPedestrian. Month names required a separate month number column to sort chronologically. A key design decision was determining which fields to include without making the dashboard exhaustive. For instance, collision trends could be shown by month or by day of the week. The monthly view was chosen because it reveals seasonal patterns relevant to policy planning, whereas the day of the week is more operational. Both light conditions and weather conditions were available, but only light conditions were included, as it directly informs infrastructure decisions like street lighting investment; weather is harder for policy to act on directly. Similarly, road surface conditions are useful for highway maintenance teams specifically, but less relevant to a broad policy analyst audience. It’s also partially correlated with the weather.

7. CONCLUSION

This report has demonstrated that Great Britain’s 2024 road collision data, when explored through an analytical dashboard, reveals clear and actionable patterns. Across 101,000 collisions and 128,000 casualties, a KSI rate of 24.84% establishes a meaningful baseline against which future years can be measured.

Single carriageways account for nearly three-quarters of all collisions, and rural roads, despite having lower volumes, carry a significantly higher KSI rate (29.60%) than urban roads (22.46%). Collision frequency peaks in June and again in October and November, reflecting the summer traffic volumes and the deterioration in light and weather conditions during autumn. The serious injury cohort, at 24,000 cases, is twelve times larger than the fatal cohort, a less visible but substantial burden. Among road users, motorcyclists and cyclists contribute to a disproportionate number of casualties, given their share of traffic, and collisions in dark, unlit conditions carry a KSI rate of 34.63%, nearly ten percentage points higher than the daylight rate.

One area needing improvement is the clarity of lighting condition categories. The distinction between “dark, no light” and “dark, lights unlit” is ambiguous, and the accompanying STATS19 dataset guide does not provide adequate differentiation between them. It can be inferred that “no light” implies an absence of all light sources, including headlights and street lighting, while “lights unlit” may indicate some surrounding ambient light, but this is not made explicit. Clearer definitions would significantly improve the reliability of any analysis involving lighting as a variable.

The findings confirm that the greatest opportunities to reduce KSI outcomes lie on single carriageways, in rural areas, among vulnerable road users, and in poor lighting conditions. Addressing ambiguities in the underlying data collection would ensure that future analysis can translate these patterns into policy with greater confidence.

8. REFERENCES

[1] Department for Transport. Road safety open data. GOV.UK. Published 25 September 2025. Available at: <https://www.gov.uk/government/statistical-data-sets/road-safety-open-data> (Accessed: 9 May 2026).